HK IMO Training 2022-2023
Problem Set 7
For 12 November 2022

Problem 25. Let $n \geqslant 4$ be an integer. Let $\mathcal{P}$ be a non-degenerate and non-self-intersecting polygon with n sides (convex or concave). For each vertex A of $\mathcal{P}$, there exists a unique vertex $f(A)$ of $\mathcal{P}$ which is closest to A. Suppose $f(A)$ is not adjacent to A for every A.

(a) Prove that $\mathcal{P}$ is a concave polygon.

(b) Find all possible $n \geqslant 4$ such that there exists such a polygon $\mathcal{P}$.

Problem 26. Let $\triangle ABC$ be a triangle with $AB < AC$. The internal angle bisector of $\angle BAC$ meets the side BC at D. Let E be the reflection of D in the midpoint of BC. Let F be a point on the internal angle bisector of $\angle BAC$ such that $EF \,/\!/\, AB$. Prove that AC is tangent to the circumcircle of $\triangle CEF$.

Problem 27. Let p be an odd prime and let n be an integer. Prove that there exists an integer m such that

$$m^{\frac{p+1}{2}} + (m+n)^{\frac{p+1}{2}} \equiv n \pmod{p}.$$

Problem 28. For any polynomial P with real coefficients, let $f(P)$ be the sum of squares of all its coefficients. For example, $f(-x^2+2x+3) = (-1)^2+(2)^2+(3)^2 = 14$. Let P, Q and R be polynomials with real coefficients satisfying $P(x)Q(x) = R(x)^2$. Prove that

$$f(P)f(Q) \geqslant f(R)^2.$$